As the Hospital Administrator, what are your five most important questions about the Emergency Department's performance?

- How many patients visited the emergency department?
- What is the average waiting time?
- Which age group visits more often?
- What percentage of patients are admitted?
- Which departments receive the most referrals?

How many rows are there? 9,216
How many columns are there? 12
Is there only one worksheet or multiple worksheets? One worksheet

Column Names
- Patient Id
- Patient Admission Date
- Patient First Initial
- Patient Last Name
- Patient Gender
- Patient Age
- Patient Race
- Department Referral
- Patient Admission Flag
- Patient Satisfaction Score
- Patient Waittime
- Patients CM

Patient Demographics	Visit Information	Operational Metrics	Administrative Information
Patient Gender	Patient Admission Date	Patient Admission Flag	Patient Id
Patient Age	Department Referral	Patient Waittime	
Patient Race			

Column Name	Description	Data Type	Category	Possible Use	Data Quality Assessment
Patient Id	Unique Identifier for each patient visit.	Text	Administrative Information	We can use it for record count and duplicates identification.	
Patient Admission	Date the patient arrived at the ED	Date	Visit Information	Allows us to	

Date				trend analysis by day, week, month, weekday, and season.	
Patient First Initial	First Initial of the Patient's name	Text	Personal Identifier	Not useful for analysis	
Patient Last Name	Patient Surname	Text	Personal Identifier	Not useful for analysis. Can only be used for Personally Identifiable Information (PII)	
Patient Gender	Male/Female/Other	Text	Patient Demographics	Help to analyse patient distribution and compare waittimes or satisfaction.	
Patient Age	Age in years	Number	Patient Demographics	Enables Age group analysis and identification of high demand groups.	
Patient Race	Ethnicity/Race	Text	Demographics	Can be used to examine equity in care and service utilizatio	

				n.	
Departme nt Referral	The department the patient was referred to	Text	Visit Information	Shows which departme nt receives the most referral and where resources may be needed.	
Patient Admission Flag	Whether the patient was admitted or not	Yes/N o	Operational Metric	Help to calculate admissio n rate	
Patient Satisfactio n Score	Patient's satisfaction rating	Numbe r	Quality Metric	Measure patient experienc e and can be compared to waittimes	Approximatel y 73% of records are blank. This likely reflects patients who did not complete a satisfaction survey rather than missing data. Blank values will be retained and excluded only from satisfaction-specific analyses.
Patient Waittime	Time before treatment	Numbe r	Operational Metric	Performa nce indicator	
Patients CM	-	-	-	-	

DATA QUALITY AUDIT

Check	Result	Decision
Unknown Column (Patients CM)	Binary (0/1), meaning unknown	Exclude until definition is confirmed
Patient Satisfaction Score	6,699 missing values; recorded scores range from 0 - 10	Keep blanks and recorded values as they are. Exclude the blanks from the

		satisfaction score analysis.
Patients IDs	No duplicates found	No action required
Patient Age	Range is 1-79 years	Values appear realistic. No cleaning required at this stage.
Patient Wait Time	10-60 minutes	Values appear realistic. No cleaning required at this stage.
Patient Gender	Values are F, M, NC	Consistent coding. Retain NC and document that its meaning is unknown until verified.
Patient Race	7 consistent categories	Valid and consistent. No cleaning required.

Finding	Result
Busiest arrival hour	11:00PM - 11:59PM
Patient records during that hour	436
Share of all 9,216 records	Approximately 4.73%
Chart Created	Patient Arrivals by Hour of Day

Average Emergency Department Wait Time by Gender
The analysis found average recorded wait times of 35.11 minutes for female patients, 35.40 minutes for male patients, and 37.17 minutes for patients in the NC category. The difference between the male and female averages was approximately 0.29 minutes. The NC category had the highest average, although its meaning and sample size require further investigation. These findings describe the observed dataset and do not establish a casual relationship between gender and waiting time.

Average Emergency Department Wait Time by Race
The analysis found that Native American/Alaska Native patients had the highest average recorded wait time at 35.69 minutes, while Pacific Islander patients had the lowest at 34.65 minutes. The difference between these averages was approximately 1.04 minutes. Average wait times were relatively similar across the seven racial categories. Further investigation of group sizes and other relevant factors is needed before drawing stronger conclusions about differences in waiting times.

The average recorded wait times were relatively similar across racial categories. However, similar averages alone do not establish that patients received equitable care. We would need additional analysis of factors such as triage severity, arrival time and other relevant characteristics.

Relationship Between Patient Wait Time and Satisfaction
The correlation coefficient between patient wait time and patient satisfaction score was **-0.0212**, indicating a negligible negative linear relationship among patients with recorded satisfaction scores. This suggests that waiting time alone has little linear association with satisfaction in the available records. However, 72.69% of patient records have missing satisfaction scores, which limits how confidently the findings can be generalized to all patients. Correlation does not establish causation.

Association Between Patient Race and Department Referral
A Chi-square test of independence was conducted to examine the association between patient race and department referral among 9,216 patient records. The results indicated a statistically significant association, $X^2(42) = 117.49$, $p<.001$. However, Cramer's V was 0.046, indicating a very weak association between the variables.

The overall referral distribution showed that 58.59% of records were classified as "None", while General Practice was the most common named referral category at 19.97%. These findings describe the observed data but do not establish that race causes differences in referral patterns.

Further investigation should consider clinical needs, patient characteristics, access to care, and the meaning of the "None" category. The findings should also be interpreted in light of the chi-square test assumptions and the quality of the underlying records.

Conclusion
This project used Excel to analyse emergency department patient data, focusing on waiting times, patient satisfaction, and department referral patterns across racial groups.

The correlation analysis showed a negligible linear relationship between waiting time and satisfaction in the analysed records. The chi-square test identified a statistically significant association between patient race and department referral, although Cramer's V indicated a very weak association.

These findings describe patterns in the available data and do not establish causation or explain why the patterns occur. Further analysis should consider clinical needs, access to care, missing data, and other relevant patient characteristics.